Glyra

A glucose digital twin with visible uncertainty

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A glucose value needs context

The review problem

A current reading shows a moment. A clinician also needs the recent trajectory, patient context and an honest view of uncertainty.

A focused use case

Explore 30–120 minute glucose forecasts for a simulated person with type 2 diabetes. Review a potential excursion before the next reading arrives.

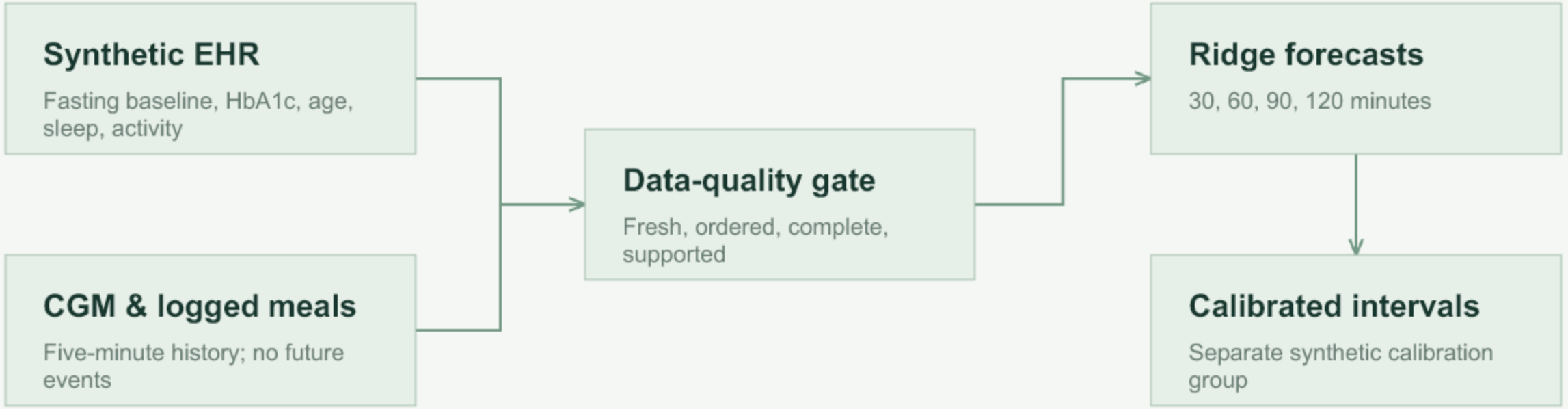
Research basis: A1C averages can conceal short glucose excursions. Long-horizon prediction remains difficult.

A working review dashboard



Synchronized history, learned forecasts, calibrated intervals and patient context

System architecture



Browser dashboard and JSON review export

Bundled model + fixtures · browser-local inference · no API keys or sensor hardware

Reproducible data and model

150 synthetic people

100 train · 25 calibration · 25 test
Four simulated days per person
27,600 total forecast windows

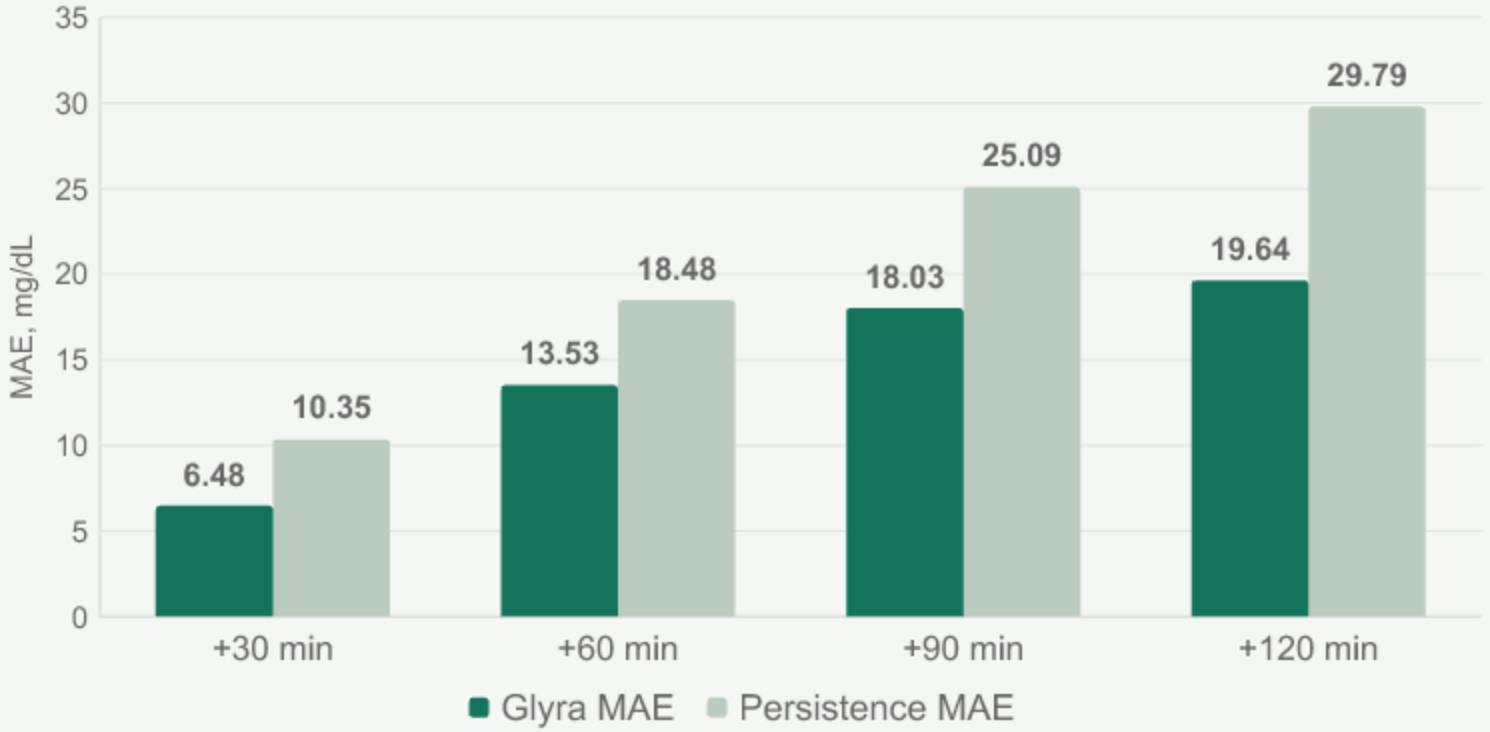
Four regression heads

Sixteen temporal and record features
Training-only standardization
Ridge penalty 8; unpenalized intercept

Generator includes variable meals, sleep, activity, sensor noise and future meals unknown to the model.

A fixed seed and separate people make the benchmark reproducible. The generator simplifies physiology.

Synthetic benchmark against persistence



19.64

mg/dL MAE at 120 min
29.79 persistence baseline

89.8% empirical coverage
90% nominal pointwise interval

25 unseen simulated people · 4,600 windows · synthetic-only performance

Explicit provenance for both streams

Static health record

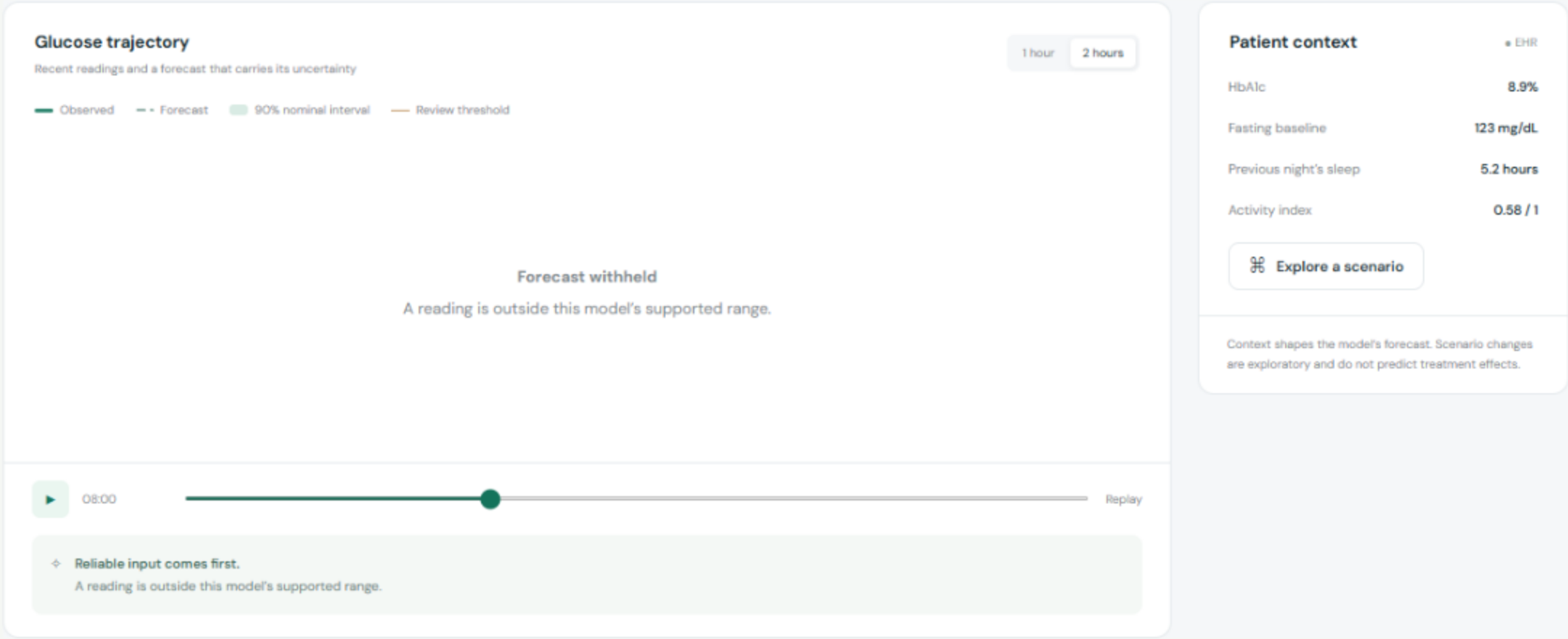
Fasting baseline, HbA1c and age accompany previous sleep and an activity index. Every patient record is labelled synthetic.

Dynamic sensor history

Five-minute glucose series and meals logged before the replay time. Future readings are excluded from features.

CSV import runs locally, validates the series and pairs it with the current demo record. No patient data leave the browser.

A missing window withholds the forecast

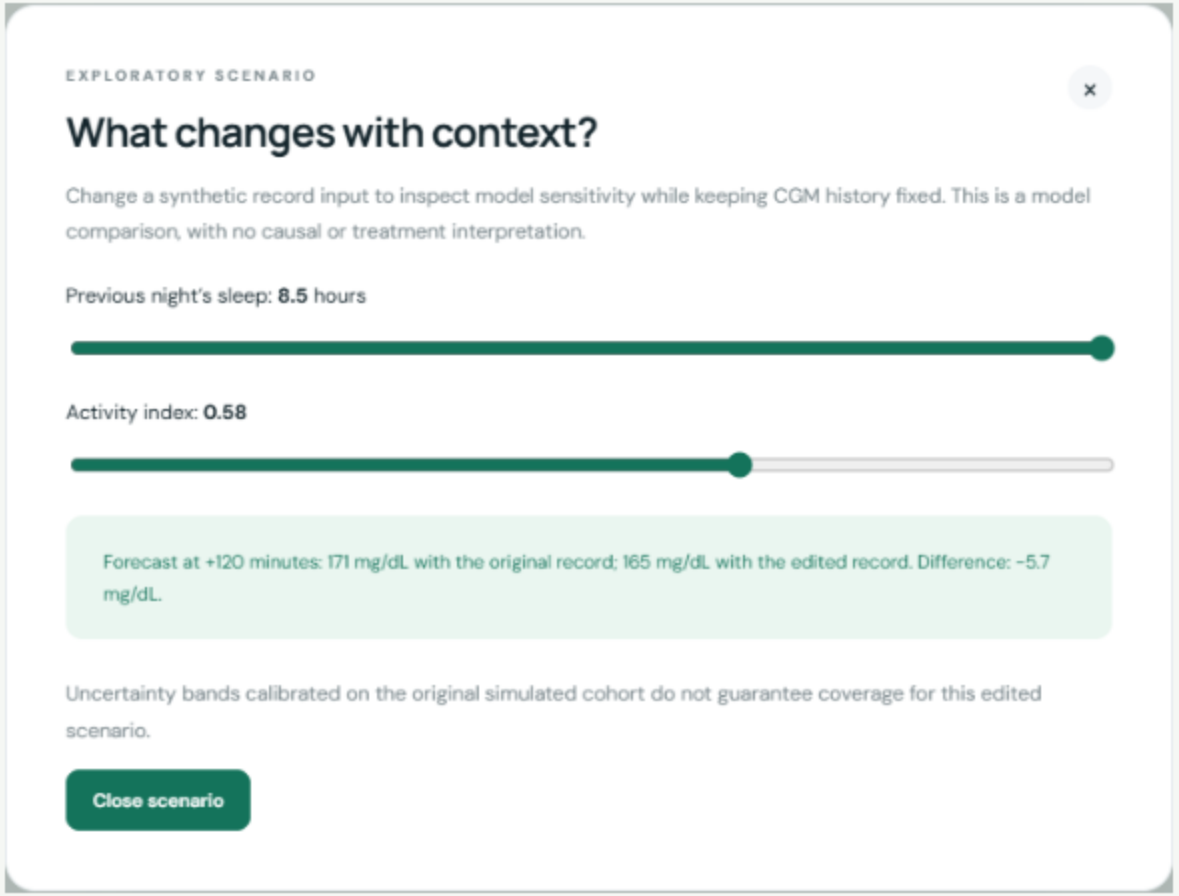


The input gate checks freshness, spacing, completeness and supported ranges before inference.

Scenario comparison exposes model sensitivity

A reviewer can adjust synthetic sleep or activity while holding the glucose history fixed.

The resulting difference describes the model. It does not establish what an intervention would do for a patient.



Limitations and failure cases

Unknown future events

An unlogged meal can change glucose after the forecast. A wide interval cannot make an unseen event predictable.

Synthetic evidence only

The generator simplifies physiology. No real patient evaluation, external validation, diagnosis or dosing capability is claimed.

Next evidence: independent retrospective testing and clinician usability review, with appropriately governed data.

Impact, adoption and implementation roadmap

Intended practical value

A review workflow could reduce the effort of reconstructing context across separate records. This remains a usability hypothesis.

Feasible next steps

Clinician interviews and task testing
Independent retrospective benchmark
A supervised research pilot

Business hypothesis: clinic subscription after validation. Pricing, demand, clinical outcomes and adoption are untested.

The current software works without hardware purchases, cloud inference or API credentials.

Source, team and reproducibility

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Browser: HTML, CSS, JavaScript
Model training: Python + NumPy
Runtime: bundled coefficients, no inference service
License: MIT

github.com/nqtexd/glyra-digital-twin